

Robayed Ashraf

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SUMMARY

AI/ML Engineer specialising in production-grade agentic AI, LLM applications, and real-time computer vision. Built DocuMind from scratch, an agentic RAG system with hybrid search, Contextual Retrieval, Redis semantic caching, and multimodal PDF ingestion, reaching 0.974 faithfulness on RAGAS evaluation. Handling everything from pipeline architecture and API development to containerisation and CI/CD, and driven by shipping things that work in the real world.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, LLM Applications, RAG, Generative AI, Deep Learning, Computer Vision, NLP, Prompt Engineering, Hybrid Search, Vector Embeddings, Contextual Retrieval, LLM Evaluation (RAGAS)

Frameworks: Python, PyTorch, TensorFlow, Keras, LangChain, LangGraph, FastAPI, OpenCV, Scikit-Learn, HuggingFace, ChromaDB, Redis

MLOps & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), ClearML, Pytest, REST API Development

Tools & Practices: Git, SQL, Agile/Scrum, Jira, Confluence

PROJECTS

DocuMind: Agentic Document Intelligence

Feb 26 – Present

Tech: LangGraph, LangChain, Gemini API, FastAPI, ChromaDB, Redis, HuggingFace, BM25, Docker, GitHub Actions, RAGAS, Pytest

- Agentic RAG pipeline with autonomous query routing, LLM-as-judge relevance grading, hallucination detection, and self-correcting retry loops.
- Hybrid retrieval combining BM25, dense vector search, RRF fusion, cross-encoder reranking, HyDE fallback, and Contextual Retrieval, improving context recall by 13.3 pp on a 30-question golden dataset.
- Rich PDF ingestion with table-aware parsing, Markdown table extraction, and Gemini multimodal figure captioning for non-text content retrieval.
- Achieved 0.974 faithfulness and 0.917 context precision; deployed with Docker Compose, Redis semantic cache, SSE streaming, and GitHub Actions CI/CD.

SignSync: Sign Language Translation for Video Conferencing

Feb 25 – Jun 25

Tech: Python, ANN, OpenCV, Gemini API (LLM integration), Computer Vision

- Trained an ANN on 3,200 images across 42 classes for real-time AUSLAN finger spelling classification, achieving 99.79% validation accuracy.
- Integrated Gemini API for LLM-based post-processing, correcting recognition errors, and improving caption readability in real time.
- Designed a 30-frame confidence-based smoothing mechanism to stabilise live predictions in continuous video streams.
- Deployed the full pipeline into a live Jitsi environment, delivering real-time sign-to-text captions as a production-viable accessibility feature.
- Presented at UTS Tech Fest AI Showcase 2025, recognised for practical impact among postgraduate AI projects.

ThirdAxis: Smart Plant Health Monitor

Feb 25 – Jun 25

Tech: Python, CNN, ClearML, Streamlit, GitHub Actions, Jira, Confluence

- Managed end-to-end ML project lifecycle using Agile practices with Jira and Confluence, covering data ingestion, preprocessing, training, evaluation, and deployment.
- Designed a modular ClearML pipeline with separate scripts for each stage, enabling reproducible and traceable experimentation.
- Implemented ClearML artifact management for dataset and model versioning, supporting experiment comparison and rollback across training iterations.
- Built a Streamlit inference interface for real-time plant disease classification from leaf images, addressing early disease detection needs in precision agriculture.

Multi-Object Tracking System for Real-Time Video Inference

Jan 25 – Jun 25

Tech: PyTorch, OpenCV, YOLOX, MOTR, TrackEval, CUDA, Streamlit

- Designed and evaluated a real-time live video tracking system, comparing MOTR (Transformer-based) and BoostTrack++ (tracking-by-detection with YOLOX) to assess feasibility for retail footfall analytics and smart city surveillance.
- Profiled inference on NVIDIA RTX 4060 (CUDA/PyTorch), recording 1.54 to 3.60 FPS and 255 to 409ms latency to identify deployment trade-offs for live video scenarios.
- Built a modular Streamlit prototype supporting webcam and live video inference for real-time visual comparison of tracking outputs across both architectures.

WORK EXPERIENCE

Self-Employed, AI/ML Engineer

Feb 26 – Present

- Sole engineer on DocuMind, leading the full product lifecycle from architecture decisions and pipeline design to deployment and iterative improvement across multiple versions.
- Built and iterated on the retrieval stack, introducing Contextual Retrieval, HyDE fallback, and Redis semantic caching based on RAGAS evaluation results.
- Designed and maintained the evaluation framework using a hand-curated 30-question golden dataset, using metric trends to prioritise and validate each improvement.
- Managed the full deployment setup, including containerisation, CI/CD pipeline, REST API design, and a pytest suite covering agent, cache, and ingestion modules.

East West University Computer Programming Club, Programming Trainer (Volunteering)

Jan 19 – Oct 21

- Mentored 50+ students in algorithms, data structures, and competitive programming with hands-on problem-solving practice.
- Designed and conducted ICPC-style mock contests, developing structured evaluation and technical communication skills.

EDUCATION

University of Technology Sydney, Master of Artificial Intelligence (Major: Computer Vision)

Aug 23 – Jun 25

WAM: 80.94 / 100 (Grade: 6.13 / 7.00)

East West University, Bachelor of Science in Computer Science and Engineering

Jan 18 – Oct 21

CGPA: 3.64 / 4.00

- Dean's List recipient and merit-based scholarship awardee for academic excellence.